

Multiword sequences do not predict speaking proficiency in dialogue: A pair-level analysis

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Abstract

This study revisited the commonly held view that the use of multiword sequences (MWSs) is a reliable predictor of L2 speaking proficiency in monologue, by extending the investigation from monologic to dialogic speaking contexts. We accessed dialogic performance data from 127 test-takers of the Test of English for Educational Purposes assessed at six different proficiency levels. The use of bigrams and trigrams was measured in terms of proportion, frequency, and association of strength. Results showed that none of the *n*-gram measures were associated with levels of speaking proficiency. When analysing pair-level use of MWSs, the results of a linear mixed-effects modeling suggested that pair-level effects explained about 40% and 30% of the total variance for bigram and trigram proportion, respectively. Follow-up analyses revealed that lower-proficiency speakers in a nationality-matched pair tended to produce a greater proportion of bigrams and trigrams than those in a nationality-unmatched pair. These findings challenge the central role of MWSs documented in monologic task performance and suggest a complex interplay of phraseological competence and speaking performance when dialogic processing is considered.

KEYWORDS

dialogic task, multiword sequence, *n*-gram, speaking proficiency

Research in multiword sequences (MWSs), word combinations that appear together frequently in a target language, has gained momentum in second language studies over the past decades. The upsurge of this line of research was initially motivated by the findings of processing advantages for MWSs in

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psycholinguistics research with both first language (L1) and second language (L2) users (e.g., Ellis et al., 2008; Siyanova-Chanturia et al., 2011; for a recent meta-analysis of MWS processing, see Yi & Zhong, 2024). Since then, the field of second language acquisition (SLA) and applied linguistics has witnessed an unprecedented interest in the role of MWSs in L2 real-life communication and has attempted to explore how and to what extent phraseological knowledge and use are associated with various aspects of L2 skills, including reading (Kremmel et al., 2017), listening (Yeldham, 2020), and writing (Kyle & Crossley, 2016). Speaking is perhaps the skill that has received the greatest attention in this line of inquiry, given the theoretical relevance of how phraseological knowledge contributes to the efficiency in oral production (for a review, see Vu & Peters, 2022). According to the modular model of speech production (Kormos, 2006; Levelt, 1989), speakers generate messages to convey (i.e., conceptualization), encode linguistic properties at lexical, grammatical, and phonological levels (i.e., formulation), and produce the utterance (i.e., articulation). It is at the formulation stage that the knowledge of MWSs is hypothesized to come into play because phraseologically competent learners benefit from the processing advantage of MWSs in retrieving and using chunks of words efficiently and fluently. These chunks allow speakers to “buy processing time while other computation proceeds” (Skehan, 1998, p. 40), and they also serve as “zones of safety” (Boers et al., 2006, p. 247), since the risk of committing errors within the chunks can be minimized, as a result, increasing speaking accuracy as well as fluency. This processing advantage hypothesis for speaking has been confirmed by the increasing number of L2 studies examining various aspects of spoken performance (e.g., Hougham et al., 2024; Kahng, 2020; Takizawa, 2024, for fluency; Uchihara et al., 2022, for lexical richness; Saito, 2020, for comprehensibility; Kyle & Crossley, 2015; Wang & Zhou, 2024, for general proficiency).

Despite the accumulated evidence supporting the role of MWSs in L2 speaking, it is important to note that all of these studies explored monologic speech. Currently, there is no evidence about whether the same relationship can be established between MWSs and proficiency development in dialogic tasks where the speakers need to interact with one another and use a range of communicative strategies (e.g., use of specific discourse markers and turn-taking) to hold a conversation. This is particularly important as previous studies (Michel, 2011; Tavakoli, 2016; Witton-Davies, 2014) have reported significant differences between monologic and dialogic speech concerning different aspects of spoken performance (e.g., fluency, syntactic complexity, and use of discourse markers).

Given the growing scholarly attention to cognitive mechanisms underlying interactional performance in L2 speaking (Peltonen, 2022; Van Os et al., 2020; Wang et al., 2018), examining whether and to what extent the central role of MWSs documented so far in monologic speaking tasks can be generalized to dialogic performance is crucial to advance our understanding of the phraseology-speaking interface in L2 speech acquisition and instruction. In addition to its importance in SLA and L2 teaching, this knowledge will be fundamental for language assessment purposes as it allows for a better understanding of what distinguishes L2 learners at different proficiency bands. Central to the assessment of speaking in a paired format test are inter-pair dynamics such as familiarity with the interlocutor, first language background and gender that are reported to affect interactional patterns between interlocutors and cause variability in test scores (O’Sullivan, 2002; Roever & Kasper, 2018; Uludag et al., 2022). Such variables and the way they may potentially affect the use of MWSs in dialogic performance have not been investigated, a gap this study aims to address.

With the rapid advancement of speech technologies and automated assessment of L2 proficiency, such knowledge will be necessary since MWSs—a set of objective, frequency-based and technology-adaptive measures, identified as a significant aspect of L2 ability—can be effectively used to evaluate L2 ability and its development (Paquot et al., 2022; Xu, 2018). Therefore, knowing how MWSs are used in a dialogic task at different proficiency levels and whether interlocutor characteristics affect that use can help develop an in-depth understanding of how MWSs are used in interaction and how such use changes as L2 learners develop their proficiency. This is the gap the current study aims to fill by examining MWSs produced in a dialogic task by 127 test-takers across different levels of proficiency.

Multiword sequences

MWS is an umbrella term used to refer to word sequences frequently occurring in a language corpus. Tavakoli & Uchihara (2020, p. 508) defined MWSs as “a catch-all term referring to combinations of words that appear together highly frequently in a target language.” Given the condition of frequent occurrence in a language corpus, MWSs naturally include a range of combinations such as collocations, phrasal verbs, idioms and binominals. They also include lexical bundles, commonly known as *n*-grams, or words that co-occur together, and they can be structurally complete (e.g., *in the end*, *on the other hand*) or incomplete (e.g., *of the*, *to be*).

In principle, there are two main approaches to examining MWSs, the phraseological approach, in which linguistic criteria (e.g., semantic transparency) are used to identify MWSs, and the frequency-based approach, in which the measurement draws on frequency of occurrence to identify recurring sequences of *n*-grams in a corpus (Gablasova et al., 2017). The frequency-based approaches can also be divided into two types: those that use text-internal comparisons and those that rely on text-external comparisons. In the former approach, the frequency of MWSs is identified within a text/corpus, for example, in a given learner corpus, by counting the frequency of consecutive strings of words occurring together. In the latter approach, frequent strings of words are first identified before they are compared against one or more reference corpora (e.g., the British National Corpus). This enables researchers to examine the extent to which the use of MWSs in the learner corpus is approaching a native-speaker baseline. The frequency-based approach to measuring MWSs (e.g., *n*-grams) is attractive to L2 speech researchers, including those interested in L2 assessment (Eguchi & Kyle, 2020), as it relies on two fundamental requirements of automated assessment of linguistic ability: Large language data (corpora) and advanced computational tools (W. Li & Liu, 2024). Following previous studies in this area (e.g., Eguchi & Kyle, 2020; Garner & Crossley, 2018), the current study has adopted the frequency-based, text-external approach to defining and measuring the use of MWSs in our test-takers' corpus. Particularly, we measure the use of *n*-grams (i.e., a sequence of *n* consecutive items from a larger sequence) exemplified by bigrams or trigrams representing contiguous sequences of two and three words, respectively.

MWSs across levels of proficiency

Several studies have examined the use of MWSs across proficiency levels in learner written corpora where samples of writing are collected as a response to an essay writing stimulus (e.g., Bestgen & Granger, 2014; Durrant & Schmitt, 2009; Granger & Bestgen, 2014; Kyle & Crossley, 2016; Laufer & Waldman, 2011; Paquot, 2019). The results of these studies suggest that as proficiency develops, L2 writers demonstrate a stronger phraseological knowledge represented by a more competent use of MWSs in terms of accuracy, range, frequency and sophistication.

Fewer studies have examined learners' phraseological knowledge in speaking and its relationship to proficiency development (e.g., Garner & Crossley, 2018; Kyle & Crossley, 2015; M. Kim et al., 2018; Tavakoli & Uchihara, 2020). In these studies, researchers investigating MWSs, typically bigrams and trigrams in terms of their key characteristics (e.g., proportion, frequency and associational strength), have reported that more proficient speakers tend to use a greater number and range of targetlike MWSs. Kyle & Crossley (2015), adopting a cross-sectional design to examine this relationship, found that oral proficiency scores based on human ratings of TOEFL speaking performances ($N = 244$) were positively associated with frequency and proportion of *n*-grams, implying that speakers at a higher proficiency level tend to use a larger number of targetlike *n*-grams that are found in the target corpus.

Similarly, a few longitudinal studies, focusing on relatively small sample sizes with dense data (e.g., 99 samples collected over one year from six L2 speakers in M. Kim et al., 2018; 214 samples collected over 4 months from 57 L2 speakers in Garner & Crossley, 2018), observed a gradual increase in both

the number of high-frequency *n*-grams and the proportion of bigrams and trigrams as proficiency developed. Tavakoli & Uchiyara (2020) examined the use of MWSs in 56 test takers' monologic speech whose proficiency was assessed, by use of their in-house standardized speaking test, at four different levels of proficiency ranging from low B1 ($n = 11$) to C1 ($n = 14$). Overall, Tavakoli and Uchiyara's findings suggested that the test-takers rated at higher proficiency levels produced a larger proportion of frequent *n*-grams and more high-frequency *n*-grams (but more weakly associated *n*-grams unexpectedly). These results in general support the commonly held view that the ability to use MWSs plays a key role in proficient L2 speech. In summary, the findings of the abovementioned studies taken together indicate that while a linear relationship between proficiency and use of MWSs is observed, the relationship is particularly reflected in both the quality of MWSs (high-frequency and strongly associated *n*-grams) and the amount of MWSs (more *n*-grams) in L2 speech.

It is worth noting that these studies have all relied on data elicited by a monologic task where the speaker performs the task individually without having to interact with an interlocutor. No study, to the best of our knowledge, has examined whether this relationship can also be observed in dialogic task performance. Examining the use of MWSs in dialogic and interactive speech is an important area of research, as previous studies have reported significant differences between speech elicited by a monologic task compared to that elicited by a dialogue.

Task effects and speaking performance

Research in task-based language teaching has provided robust evidence to suggest that task design influences task performance in a number of different ways (Robinson, 2011; Skehan, 1998; Wright, 2021). Several design features, for example, task structure (i.e., the degree to which task components could be re-organised), have been shown to promote accuracy and fluency of speech (Foster & Tavakoli, 2009); other design features, such as task type, are reported to affect syntactic and lexical complexity of the language elicited (Tavakoli & Foster, 2008). One aspect of task design frequently reported to have an impact on spoken performance is the speaking mode, whether the task is performed individually as a monologue or interactively with an interlocutor in the form of a dialogue.

Michel (2011), using a within-participant design, examined L2 learners' speaking performance in the same task twice, once as a monologue and once while performing it as a dialogue. The monologic task involved leaving a message on a friend's answering machine, whereas the dialogic task involved giving the message to a friend on the phone while explaining and discussing it. The results suggested that dialogic speech was more fluent than that in a monologue. Michel's (2011) findings also suggested interaction promoted accuracy and enhanced, although to a limited extent, lexical complexity. Other studies have also reported similar differences between monologic and dialogic task performance in terms of fluency (e.g., Huensch, 2023; Peltonen et al., 2025; Tavakoli, 2016; Witton-Davies, 2014), syntactic complexity (Tavakoli, 2018) and use of discourse markers (Maragianni, 2023), with dialogic performance being more fluent and complex, usually eliciting a more frequent use of discourse markers. Examining the effects of L1 speaking style and task mode in a monologic versus dialogic task in a within-participant design, Peltonen et al. (2025) reported different task mode effects on different dimensions of fluency. The authors explained the differences in light of "the presence of the interlocutor and the overall interactive nature of the dialogic task" (Peltonen et al., 2025, p. 17).

When explaining task mode effects, some researchers (Michel, 2011; Peltonen, 2017; Peltonen et al., 2025; Tavakoli, 2018) attribute the improvements in dialogic performance (e.g., in fluency) to the nature and organisation of dialogic task performance in which speakers "can use the interlocutor's turn to plan their own subsequent performance" (Tavakoli, 2018, p. 136). Other researchers (e.g., Huensch, 2023, p. 1034), on the other hand, argue that performing a monologic task is less demanding "in terms of both joint attention and inferring social intentions," underlining the demands associated with turn-taking and interaction management. Regardless of whether they make task performance more or less demanding, Peltonen et al. (2025) argue that task mode effects on L2 speech are more complex than

previously anticipated, as they dynamically interact with other learner- and context-related factors (e.g., L1 background). They call for task mode effects to be carefully examined and accounted for if an in-depth understanding of L2 production process and a reliable assessment of L2 proficiency are intended. In line with this argument and highlighting the differences between monologic and dialogic speech, Gao & Sun (2025, p. 1) contend that language tests may need to consider adjusting their rating rubrics “to account for differences between monologic and dialogic speaking.” The brief review of task mode effects above underlines the crucial role of task mode and its effects on performance, an area that has not been investigated in studies examining L2 phraseological competence.

Interlocutor effects

Interlocutor effects, or the way the interlocutor’s characteristics affect test-takers’ performance, have been investigated in language assessment. These effects are crucially important as they can mask candidates’ true language ability, encouraging construct-irrelevant variance in test scores and leading to less valid and reliable test results. Previous research (Davis, 2009; Jaiyote, 2015; McNamara & Lumley, 1997; O’Sullivan, 2002) has shown that interlocutors’ proficiency level, acquaintanceship, gender and L1 background are some of the factors potentially affecting language performance in paired speaking tasks. For proficiency effects, Davis (2009) reported more language production when candidates are paired with a higher proficiency level interlocutor, maintaining that unless candidates are paired with an interlocutor at a much lower proficiency level, interlocutor effects of proficiency are limited. Examining familiarity with the interlocutor, O’Sullivan (2002) reported evidence to suggest that speaking to someone the test-takers are familiar with can help them achieve higher scores, with positive effects on the complexity of the language used. Relatedly, a mixed-methods study by Jaiyote (2015) indicated that speakers who share the same L1 improve mutual understanding by drawing on their cultural and linguistic backgrounds, even when what is said is unclear or ambiguous. Findings of research in relation to gender effects are inconclusive, as some studies (e.g., O’Sullivan, 2000; Porter, 1991) reported effects of gender on performance, while others (e.g., O’Loughlin, 2002) did not find strong evidence of an interlocutor gender effect on test results across candidates of different nationalities. Using Rasch Analysis to examine the interaction of test-taker gender, task topic and interlocutor gender, Lumley and O’Sullivan (2005), reported small effects for some, but not all, of the interactions. The authors concluded there was limited evidence for the effects of gender on performance on its own or in its interaction with the topic of conversation.

The current study

The review of the literature presented above suggests that MWSs have been researched extensively in L2 writing and to a lesser extent in L2 speaking in monologic tasks, and in both sets of studies, a linear relationship has been observed between the use of MWSs (i.e., bigrams and trigrams) and proficiency development. The current study aims to help expand our understanding of the use of MWSs in a dialogic task across a range of proficiency levels. It further examines the degree of variability in MWS use across pairs and whether such variability can be accounted for by three pair-level characteristics—gender, nationality and proficiency—which have been extensively investigated in the paired speaking assessment literature (e.g., Davis, 2009; Jaiyote, 2015; McNamara & Lumley, 1997; O’Sullivan, 2002).¹ By assessing proficiency in a standardized and validated test of English, our study will provide valuable insight into the role MWSs play in L2 speaking proficiency. The following research questions (RQs) guide our study.

RQ1. To what extent does the use of MWSs relate to assessed levels of proficiency measured through a dialogic speaking task?

TABLE 1 TEEP speaking paper format.

Part	Task	Mode	Example	Planning time	Response time
0	Pre-viewing the focus question/topic introduction	Silent preparation	<i>Question: Which is better; private or public services?</i>	20 s	-
1	Individual talk (role plays)	Monologue	<i>The advantages of publicly funded services.</i>	4 min	3 min
2	Scenario discussion	Dialogue	<i>Discuss with your partner and analyse the question</i>	2 min	4 min
3	Focus question discussion	Dialogue	<i>Discuss the focus question with your partner and agree or disagree!</i>	None	2 min

- RQ2. To what extent does the use of MWSs vary across pairs?
- RQ3. How do pair-level characteristics (gender, nationality and proficiency) influence the use of MWSs?

METHOD

The study used the data from the Test of English for Educational Purposes (TEEP), an academic English proficiency assessment developed at the University of Reading for the purpose of evaluating the readiness of international students for study in English-medium higher education. TEEP includes five different components of language knowledge: speaking, writing, listening and reading. TEEP scores are reported on a 0–9 scale with half-band increments. In typical admissions practice, a score of 5.0 or above may meet foundation-level entry requirements, while 6.0 or above is generally required for undergraduate or postgraduate study (for more detailed information about the test, see <https://www.reading.ac.uk/global-academy/english-language-tests/teep>).

The TEEP speaking test is administered in a paired format and includes three interrelated tasks eliciting different kinds of spoken performance on the same topic/question (see Table 1 below). TEEP speaking test topics do not require specialist subject knowledge and center around common societal issues such as public services or public education.

The TEEP speaking is rated on a 9-point scale, with zero representing a speaker who makes no attempt to speak and nine indicating a proficient speaker.² Levels 3 and below are defined as “limited speaker[s],” and Level 9 is defined as a “very good speaker.” The rating scales use both global and analytic rating scales for these criteria: *explaining ideas and information, interaction, fluency, accuracy + range of vocabulary and grammar and intelligibility*. Two trained examiners assess the speaking performance, in which one is an interlocutor (i.e., facilitating speaking tasks using a standardized script and rating candidates’ performances on two of the assessment criteria) and the other an assessor (i.e., sitting quietly at the back of the room while listening and awarding grades on all five criteria). Using the set of validated marking scales and marking descriptors for each of the five criteria mentioned above, the assessors and interlocutors rate the candidate’s performance. The scores given by the examiners are averaged, and the inter-rater reliability is calculated based on the explanation of ideas and interaction sub-scores awarded by both raters. Internal test consistency is regularly monitored to ensure all criteria are distinguishing between proficiency levels reliably. A comparative study of TEEP and IELTS (D. Li et al., 2025) indicates that, although both tests employ comparable nine-point rating scales, TEEP speaking scores tend to be slightly higher than IELTS speaking scores when administered to the same test-takers. Nevertheless, the study reports that both tests appear to measure the same underlying language construct, namely “English language proficiency” (p. 228).

matched pairs tended to produce a greater proportion of sequences when compared to nationality-unmatched pairs.

- The effect of nationality was pronounced for lower-proficiency learners, meaning that lower-proficiency speakers (ranging somewhere between 5.0 and 6.0) used a greater proportion of bigrams and trigrams when they were in the nationality-matched pairs.

Things to consider

- This study challenges the prevailing view that multiword sequence use serves as a dependable indicator of advanced speaking proficiency.
- Speech production process may be different between monologic and dialogic performance. In the latter, when people are having a conversation, they naturally start to mirror each other's language, using similar words, phrases, and sentence patterns (i.e., linguistic alignment). Thus, much of the "thinking work" needed to speak fluently is already done while listening to the other person. Because of this, using ready-made phrases (multiword sequences) doesn't give speakers much extra advantage in being fluent during a conversation.
- In general, teachers and assessors should not rely on the use of multiword sequences as a useful indicator of speaking proficiency when speech is elicited in a paired format.
- Yet, teaching multiword sequences may help low-proficiency learners to improve their speaking skills in a dialogic context.

Material, data, open access article: Data and open access paper available from <https://osf.io/eyx23>

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FIGURE 1 A sample TEEP speaking scenario task. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

Target tasks and topics

The focus in this study is on the scenario discussion task used for rating interactional skills. The candidates are given a visual prompt with a problem and three possible solutions (see Figure 1 for a sample TEEP speaking scenario task). The task prompts candidates to discuss topics in a role rather than from a personal perspective. The role might be as a member of a particular community or wider society in order to simulate the target academic context and to elicit more abstract/complex language use. Following 2 min of preparation time, candidates are allowed 4 min to discuss the scenario and evaluate the options. Provided with paper and pencils, candidates are allowed to take notes during preparation and may refer to them during the discussion. In practice, candidates take more notes during the monologic task, which allows 4 min of preparation. In the dialogic task (the focus of this study), the 2-min preparation time is typically used to read the scenario, plan responses and ask the interlocutor for clarification of unfamiliar vocabulary, which is an allowed and encouraged practice that is not penalized to prevent lexical limitations from affecting interactive performance. Cases of candidates using long-scripted answers are rarely observed, likely due to the limited preparation time intended to discourage extensive note-taking. Candidates A and B receive the same prompt for this task and are given explicit instructions to manage the interaction on their own and to collaborate.

Speaking data

Data came from the dialogic performance of 127 candidates (72 females and 55 males) taking the TEEP speaking paper before starting their university degree. The sample included 118 postgraduate and 9 undergraduate candidates from a range of ethnic, linguistic and disciplinary backgrounds, including Chinese ($n = 65$), Saudi ($n = 22$), Kazakh ($n = 15$), Thai ($n = 11$), Brazilian ($n = 4$), Iraqi ($n = 2$), and one participant each for the remainder (Libyan, French, Ghanaian, Indian, Jordanian, Omani, Turkish and Ukrainian). Learners were paired with familiar partners of a similar level of ability by their class tutors, in line with O'Sullivan's (2002) observations on the positive effects of interlocutor familiarity. The study took a purposeful sampling approach, selecting performances across a range of proficiency levels from a database of TEEP speaking audio recordings made between 2014–2019. To control for rater bias as a variable, the sample was narrowed down to the recordings rated by examiners with high inter-rater reliability indices (intraclass correlation coefficient single and average measures >0.70). The final sample included 127 performances rated at TEEP levels of 5.0 ($n = 20$), 5.5 ($n = 21$), 6.0 ($n = 19$), 6.5 ($n = 21$), 7.0 ($n = 25$) and 7.5 ($n = 22$).³ Of the full set of individual

samples ($N = 127$) used for analysis to answer RQ1, 92 speaking samples (46 pairs) were available for pair-level analyses to answer RQs 2 and 3. The remaining 35 individual samples were elicited in a paired assessment format, but the samples from their interlocutors were not available in the current data set due to issues with data quality (e.g., not sufficiently audible) and purposeful sampling (i.e., to obtain sufficient samples in each level). Among 46 pairs, 22 pairs were matched on proficiency (48%), 23 on gender (50%) and 21 on nationality (46%).

The candidates' performances were initially transcribed verbatim by an experienced research assistant while listening to the conversations, often several, and transcribing each speaker's speech while identifying turns taken by each. Incomprehensible or ambiguous pronunciations identified by the research assistant were checked by one of the researchers. Any unclear or incomprehensible pronunciations, although very few, were marked as "X" in the transcription. Empty fillers (e.g., *uh* and *um*) and non-linguistic sounds (e.g., laughter) were removed for the purpose of n -gram analysis, but all repair strategies (e.g., repetition or reformulation) were kept in the transcriptions. A sample of 20% of the transcriptions was checked by one of the authors, and an agreement of 96% was established for the accuracy of the transcriptions.

Measuring MWSs: n -gram assessment

Building on prior phraseological studies (e.g., Garner & Crossley, 2018; Tavakoli & Uchihara, 2020), we utilized eight n -gram indices—proportion (two measures), frequency (two measures) and association (four measures)—based on two- and three-word continuous sequences (i.e., bigrams and trigrams). To generate the n -gram scores, we employed the Tool for the Automatic Analysis of Lexical Sophistication 2.2 (TAALES: Kyle & Crossley, 2015; Kyle et al., 2018). TAALES is a modern computational tool designed to evaluate multiple dimensions of lexical sophistication in learner language, including word frequency, distribution and psycholinguistic properties. For our reference corpus, we chose the spoken component of the Corpus of Contemporary American English (COCA: Davies, 2009), due to its substantial size. This subsection contains 79 million words sourced from spontaneous dialogues in a diverse array of U.S. television and radio programs recorded over the past 25 years. The three types of bigram and trigram measures (proportion, frequency and association) were computed for each of 127 individual texts. In what follows, the description of the three types of n -gram measures (proportion, frequency and association) is provided.

n -gram proportion

Proportion score indices represent those of bigrams and trigrams in learner speech that also appear in a reference corpus. In this study, TAALES measured the proportion of bigrams and trigrams from learner samples that matched the 30,000 most frequent n -grams in the spoken section of COCA. A higher proportion score reflects a greater use of target-like n -grams, with the total number of n -grams in text taken into account. Prior studies have shown that advanced L2 learners tend to produce a higher proportion of such n -grams in their spoken output (Garner & Crossley, 2018; Kyle & Crossley, 2015).

n -gram frequency

In contrast to n -gram proportion indices, which rely on binary scoring (i.e., whether or not learner-produced n -grams appear in the reference corpus), frequency score indices operate on a continuous scale. Each bigram and trigram in a learner's text is given a frequency score, and these are averaged to produce a single overall score for each speaker. We used logarithmic frequency scores for bigrams and trigrams rather than raw frequencies to account for Zipfian distributions typically seen in word fre-

quency data. Higher frequency scores suggest that learners produce more high-frequency, target-like n -grams, while the total number of n -grams in text with frequency score was accounted for. Previous research indicates that more proficient L2 speakers tend to use a greater number of high-frequency n -grams in their speech compared to less proficient learners (Garner & Crossley, 2018; Kyle & Crossley, 2015).

n -gram association

n -gram association score indices reflect the strength of the association between individual words within bigrams or trigrams. While association indices are similar to proportion and frequency indices in that they all rely on word co-occurrence frequencies from a reference corpus, they differ by minimizing the influence of the individual word frequencies. This is achieved by factoring out the frequencies of individual words when calculating association scores. In this study, we selected two association measures—mutual information (MI) and delta P (DP)—which have been widely applied in SLA studies examining MWSs in both spoken and written contexts (e.g., Durrant & Schmitt, 2009; Eguchi & Kyle, 2020; Garner & Crossley, 2018; Granger & Bestgen, 2014). Both indices compare the actual frequency of n -grams in the reference corpus (COCA, in this case) to the expected frequency based on the individual word frequencies (Gablasova et al. 2017). One major difference between these two measures is that DP accounts for directionality (Gries & Ellis, 2015), or the asymmetrical nature of co-occurrence of individual words. For instance, the degree to which we can predict the occurrence of the word *circumstances* is much higher when first hearing *extenuating* than the other way around, given that *circumstances* can open up a myriad of candidates for partnership that can go with the word, such as *difficult*, *challenging*, *medical*, *current* and *historical*. We adopted trigram measures reflecting unigram to bigram associations given that such measures represent the construct of trigram association strength (e.g., DP, Eguchi & Kyle, 2020). Prior research has shown that high-proficiency L2 writers tend to use more sophisticated combinations with higher MI and DP scores compared to lower-proficiency writers (Bestgen & Granger, 2014; Durrant & Schmitt, 2009; Granger & Bestgen, 2014; Kyle et al., 2018). However, the exact nature of the relationship between these association measures and spoken language proficiency remains unclear, with evidence showing null findings (Garner & Crossley, 2018; M. Kim et al., 2018), or the relationship can be positive (i.e., higher MI and higher proficiency, Hougham et al., 2024) or negative (i.e., higher MI and lower proficiency, Tavakoli & Uchihara, 2020).

RESULTS

RQ1: To what extent does the use of MWSs relate to assessed levels of proficiency measured through a dialogic speaking task?

Descriptive statistics and details of the results of statistical analyses to answer the three research questions can be found in Appendices 1 to 5 via OSF (<https://osf.io/eyx23>).

To answer RQ1, we first conducted a series of correlation analyses between speaking proficiency and eight n -gram measures. Because speaking proficiency was considered an ordinal categorical variable, we used a non-parametric test (i.e., Spearman's rank correlation). Table 2 shows that none of the n -gram measures, except for two (trigram proportion and DP), significantly correlated with speaking proficiency. The correlations for the two trigram measures, however, were small and almost negligible ($r < 0.30$). We also conducted an analysis of variance (ANOVA) with the proficiency level as the independent variable (5.0, 5.5, 6.0, 6.5, 7.0 and 7.5 levels) and with each of eight n -gram measures as the dependent variable. Because we confirmed that the assumption of equal variances for some n -gram measures was violated (see Appendix 2 for the results), we adopted a Welch's F -test, while correction

TABLE 2 Spearman's rank correlation between *n*-gram measures and speaking proficiency.

	<i>r</i>	<i>p</i>
<i>Bigram measures</i>		
Proportion	0.095	.288
Frequency	0.105	.239
MI	0.120	.178
DP	0.068	.449
<i>Trigram measures</i>		
Proportion	0.182*	.041
Frequency	0.138	.122
MI	0.148	.096
DP	0.281**	.001

Note: **p* < .05; ***p* < .01.

for unequal variances was applied. The analysis showed that none of the *n*-gram measures significantly distinguished different levels of speaking proficiency (*p* = .111 to .869), consistent with the results of correlation analysis (see Appendix 2 for the results; see Figures 2 and 3 for the plots). These results indicate that the use of MWSs was not distinctive across L2 learners with different speaking proficiency levels when their speech was elicited via a dialogic speaking task.

RQ2: To what extent does the use of MWSs vary across pairs?

In answer to RQ2, we focused exclusively on the data set available at the pair level. As a result, the subsequent analyses further examined the data from a total of 46 pairs (*N* = 92: *n*_{5.0} = 9, *n*_{5.5} = 12, *n*_{6.0} = 16, *n*_{6.5} = 13, *n*_{7.0} = 20, *n*_{7.5} = 22). A linear mixed-effects model was fit to the data using restricted maximum likelihood (REML) estimation with proficiency level as the fixed effect and by-pair intercept as the random effect, predicting each of the eight bigram and trigram measures. Because our primary interest was in the degree of pair-level variation in the MWS use, we first produced the intraclass correlation coefficient (ICC) for the by-pair random effect (maximum range = 0 to 1). The ICC was calculated by dividing the variance of the random effect (i.e., by-pair intercept) by the total random variance (i.e., sum of the by-pair random effect and error). The larger ICC indicates that the larger variation in the use of MWSs can be uniquely explained by the pair-level effect and suggests the greater similarity in the MWS use between speakers in the same pair. In addition, we conducted a likelihood ratio test (LRT) for the random effect to examine whether the pair-level effect would be a significant factor in the model. The LTR compares the likelihood of a model with the pair-level effect included versus a model with the effect excluded.

The analysis of the ICC showed that for the two proportion measures, relatively higher ICCs were observed for proportion measures (0.421 for bigram proportion; 0.299 for trigram proportion), while the ICC for other measures was negligible, ranging from 0.055 to 0.180 (see Appendix 3 for the results). These results mean that approximately 40% and 30% of the total variance for bigram and trigram proportion, respectively can be explained by the pair-level effect alone. The LRT results showed that the pair-level effect was significant for bigram proportion, LTR(1) = 8.45, *p* = .004 and approached significance for trigram proportion, LTR(1) = 3.55, *p* = .060. Indeed, we found a significant (or marginally) speaker-interlocutor correlation for the proportion measures (bigram: *r* = 0.404, *p* = .004; trigram: *r* = 0.259, *p* = .076), indicating that the greater the number of MWSs (especially for two-word sequences) one speaker used in their speech, the greater the number of MWSs the partner in the same pair used (see Appendix 4 for the results of other *n*-gram measures). These findings indicate

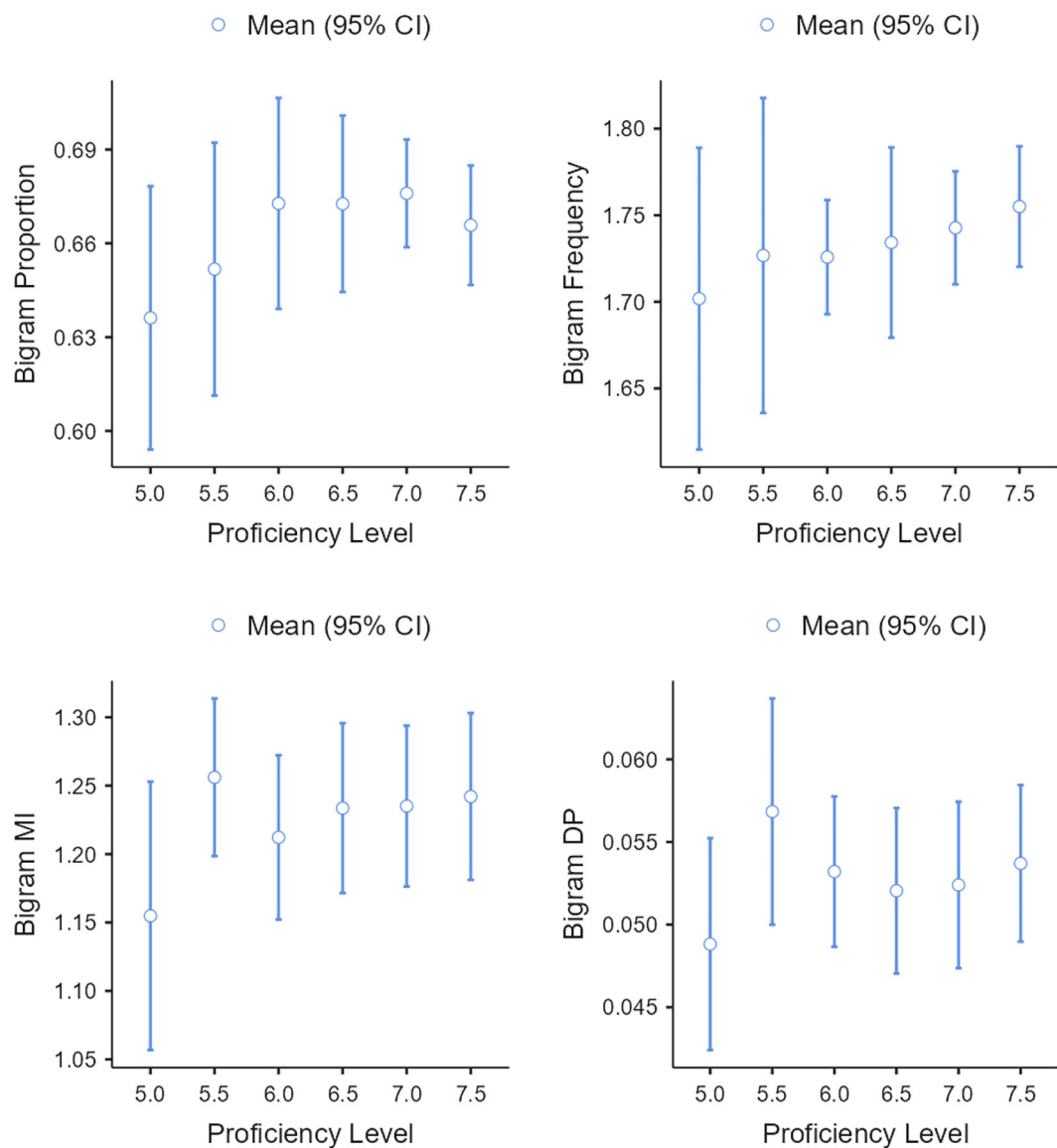


FIGURE 2 Bigram measures and speaking proficiency levels. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

that there was an important between-pair variation and a within-pair similarity in the proportion of MWSs produced between two speakers in the same pair.

RQ3: How do pair-level characteristics (gender, nationality and proficiency) influence the use of MWSs?

In answer to RQ2, we found there was a significant pair-level variation in the proportion of MWSs used. To answer RQ3, we attempted to further reveal the extent to which such between-pair variability in n -gram proportion measures can be explained by the three pair-level variables (gender, nationality and proficiency). Prior to the analysis, we created pair-level variables for each of the three characteristics regarding whether the gender, nationality and proficiency of the two speakers in the same pair

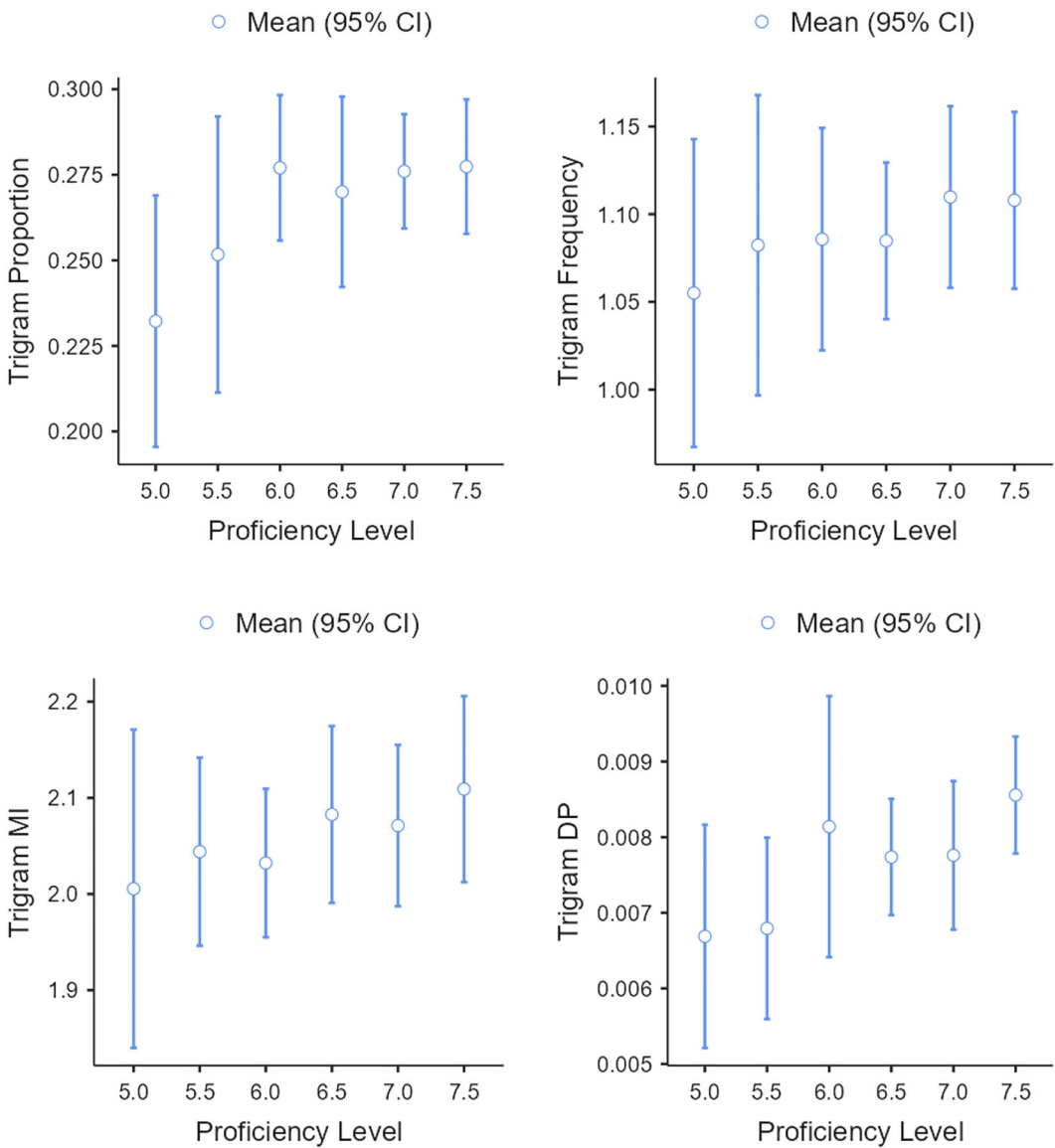


FIGURE 3 Trigram measures and speaking proficiency levels. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

were matched. As a result, dichotomous, dummy-coded data were created for 46 pairs for proficiency (number of matched pairs = 22), gender (number of matched pairs = 23) and nationality (number of matched pairs = 21). Then, a linear mixed-effects model was fit to the data using REML estimation with the three pair-level variables as the fixed effects predicting each of the bigram and trigram proportion measures.

As presented in Tables 3 and 4, pair-level nationality significantly explained the variance of the bigram proportion ($b = 0.036, p = .028$), and a similar trend was observed for the trigram proportion ($b = 0.028, p = .052$). These results indicate that when the nationality of the two speakers in the same pair was matched, they tended to produce a greater proportion of bigrams and trigrams in speech. In contrast, neither gender nor proficiency showed a significant effect in the models for bigram or trigram proportions.

TABLE 3 Mixed-effects model with pair-level characteristics (matched vs. unmatched) as predictors of bigram proportion.

Fixed effects	<i>b</i>	95% Confidence interval		<i>t</i>	<i>P</i>
		Lower	Upper		
Intercept	0.659	0.628	0.690	41.85	<.001
Gender	−0.004	−0.034	0.026	−0.25	.804
Proficiency	−0.019	−0.050	0.012	−1.22	.230
Nationality	0.036	0.005	0.067	2.28	.028
Random effects	<i>SD</i>	Variance	ICC		
By-pair intercept	0.038	0.001	0.380		
Residual	0.049	0.002			

Note: Marginal $R^2 = 0.117$; Conditional $R^2 = 0.453$.

TABLE 4 Mixed-effects model with pair-level characteristics (matched vs. unmatched) as predictors of trigram proportion.

Fixed effects	<i>b</i>	95% Confidence interval		<i>t</i>	<i>p</i>
		Lower	Upper		
Intercept	0.260	0.232	0.387	18.46	<.001
Gender	0.003	−0.024	0.030	0.23	.822
Proficiency	−0.017	−0.045	0.011	−1.20	.238
Nationality	0.028	0.001	0.056	2.00	.052
Random effects	<i>SD</i>	Variance	ICC		
By-pair intercept	0.031	0.001	0.294		
Residual	0.048	0.002			

Note: Marginal $R^2 = 0.094$; Conditional $R^2 = 0.361$.

Exploratory analysis

Given that nationality significantly influenced the use of MWSs in a dialogic task, we speculated that the relationship between speaking proficiency and *n*-gram proportion measures might have been modulated by the pair-level nationality. To further explore the possible reason for the lack of a significant link between L2 proficiency and MWS use in answer to RQ1 (cf. Tavakoli & Uchihara, 2020), we conducted a follow-up mixed-effects regression analysis. In this model, we entered two predictor variables as the fixed effects (nationality and proficiency level) and their two-way interaction term for predicting each of the bigram and trigram proportion measures. The proficiency variable was dummy coded, and Level 5.0 was set as a reference level.

As summarized in Tables 5 and 6, there were significant interactions between proficiency level and nationality for bigram and trigram proportion measures. Figure 4 demonstrates that for nationality-unmatched pairs, there seems to be a positive and linear relationship between the *n*-gram proportion and proficiency from Level 5.0 up to around Level 6.5, while for nationality-matched pairs, such a positive and linear relationship does not seem to emerge. For the bigram proportion, for instance, the marginal mean for matched pairs showed a gradual increase across proficiency levels, rising from 0.58 at Level 5.0 to 0.60 at Level 5.5, 0.64 at Level 6.0, and reaching 0.69 at Level 6.5. In contrast, the marginal mean for unmatched pairs remained relatively stable across levels, staying close to 0.70 (0.72 at Level 5.0, 0.69 at Level 5.5, 0.70 at Level 6.0, and 0.69 at Level 6.5). This

TABLE 5 Mixed effects model with pair-level nationality and proficiency as predictors of bigram proportion.

Fixed effects	<i>B</i>	95% Confidence interval		<i>t</i>	<i>p</i>
		Lower	Upper		
Intercept	0.583	0.526	0.641	19.96	<.001
Nationality	0.134	0.053	0.214	3.26	.002
Level 5.5	0.017	−0.054	0.087	0.47	.643
Level 6.0	0.059	−0.014	0.131	1.58	.119
Level 6.5	0.103	0.027	0.179	2.65	.010
Level 7.0	0.090	0.021	0.158	2.58	.013
Level 7.5	0.078	0.011	0.145	2.28	.027
Nationality × Level 5.5	−0.041	−0.142	0.061	−0.79	.432
Nationality × Level 6.0	−0.072	−0.171	0.026	−1.44	.153
Nationality × Level 6.5	−0.129	−0.232	−0.027	−2.48	.015
Nationality × Level 7.0	−0.138	−0.236	−0.040	−2.77	.007
Nationality × Level 7.5	−0.124	−0.220	−0.028	−2.53	.014
Random effects	<i>SD</i>	Variance	ICC		
By-pair intercept	0.033	0.001	0.317		
Residual	0.049	0.002			

Note: Marginal $R^2 = 0.247$; Conditional $R^2 = 0.486$.

diverging pattern was particularly salient for lower-proficiency speakers (Levels 5.0 and 5.5), with 95% confidence intervals not or barely overlapping between speakers in nationality-matched pairs and those in nationality-unmatched pairs (see Appendix 5 for the results of estimated marginal means and confidence intervals). These findings suggest that when two lower-proficiency interlocutors shared the same L1 and cultural background, they were more likely to produce a greater proportion of MWSs. On the other hand, when their L1 and cultural background were not shared, such an inflation of the use of MWSs may not be observed (particularly for beginning-intermediate L2 learners ranging from Level 5.0 up to around 6.0 to 6.5); hence, the *n*-gram proportion measure may serve as an indicator of L2 speaking proficiency in this group.

Discussion

The present study examined the relationship between the use of multiword sequences (MWSs) and oral proficiency when it was measured in a paired speaking task format. Addressing this question extends our understanding of the phraseology-speaking link, predominantly based on the analysis of monologic speaking performance, to provide further insights into the role of MWSs in real-life interactions and oral communication. The results in answer to RQ1 (i.e., the relationship between *n*-gram measures and oral proficiency) showed that none of the *n*-gram measures were related to speaking proficiency scores in TEEP, a finding that is inconsistent with previous research based on monologic speaking tasks (e.g., Tavakoli & Uchihara, 2020). This finding implies that while MWS use is known to reflect proficiency and distinguish levels of proficiency in monologic task performance, the use of MWSs, whether in proportion, frequency or association, is not a reliable indicator of proficiency in a dialogic task. The interactional nature of a dialogic task and the potential impact of the interlocutor’s characteristics can perhaps explain the findings (see further discussion below).

The results in answer to RQ2 (i.e., variation in the use of *n*-grams across pairs) demonstrated a large pair-level variation in the proportion of *n*-grams relative to an individual-level variation. These results

TABLE 6 Mixed effects model with pair-level nationality and proficiency as predictors of trigram proportion.

Fixed effects	<i>B</i>	95% Confidence interval		<i>t</i>	<i>p</i>
		Lower	Upper		
Intercept	0.182	0.134	0.230	7.46	<.001
Nationality	0.122	0.052	0.191	3.42	.001
Level 5.5	0.021	−0.039	0.081	0.69	.495
Level 6.0	0.073	0.012	0.134	2.34	.023
Level 6.5	0.077	0.012	0.143	2.31	.024
Level 7.0	0.097	0.039	0.154	3.30	.002
Level 7.5	0.095	0.039	0.151	3.32	.002
Nationality × Level 5.5	0.007	−0.084	0.099	0.16	.877
Nationality × Level 6.0	−0.082	−0.168	0.004	−1.86	.067
Nationality × Level 6.5	−0.108	−1.99	−0.018	−2.35	.021
Nationality × Level 7.0	−0.140	−0.224	−0.056	−3.26	.002
Nationality × Level 7.5	−0.121	−0.204	−0.038	−2.85	.006
Random effects	SD	Variance	ICC		
By-pair intercept	0.020	0.000	0.146		
Residual	0.048	0.002			

Note: Marginal $R^2 = 0.296$; Conditional $R^2 = 0.399$.

suggested that 42.1% of the total variance in the bigram proportion and 29.9% of the variation in trigram proportion were attributable to differences between pairs. The speaker-interlocutor correlation also showed that the more bigrams a speaker produced, the more bigrams the interlocutor was likely to use ($r = 0.404$, $p < .01$). This finding provides evidence to support the emergence of linguistic alignment at a lexical level in a paired speaking task (e.g., Michel et al., 2022; Pickering & Garrod, 2021).

Lastly, the results in answer to RQ3 (i.e., the impact of pair-level features on n -gram measures) showed that of the three pair-level variables (gender, nationality and proficiency level), nationality was the only variable related to n -gram proportion measures. This indicated that speakers in the nationality-matched pairs tended to produce a greater proportion of n -grams when compared to nationality-unmatched pairs. A follow-up analysis also showed that lower-proficiency speakers (ranging somewhere between 5.0 and 6.0) used a greater proportion of bigrams and trigrams when they were in the nationality-matched pairs, implying they benefited from being paired with someone from their national background.

In what follows, we address two questions arising from the current analysis that merit further exploration in order to advance the understanding of the relationship between phraseological competence and speaking proficiency: (a) why did none of the n -gram measures relate to speaking proficiency in a dialogic task? and (b) why were nationality-matched pairs able to produce a greater proportion of n -grams, particularly when their L2 oral proficiency was within the lower range?

Why is the use of MWSs not a reliable predictor of L2 proficiency in dialogue?

Although we predicted that the use of MWSs indicated by n -gram measures would be significantly associated with speaking proficiency, the current findings turned out to be contradictory. This prediction was based on the view that phraseologically competent speakers, compared to those with limited

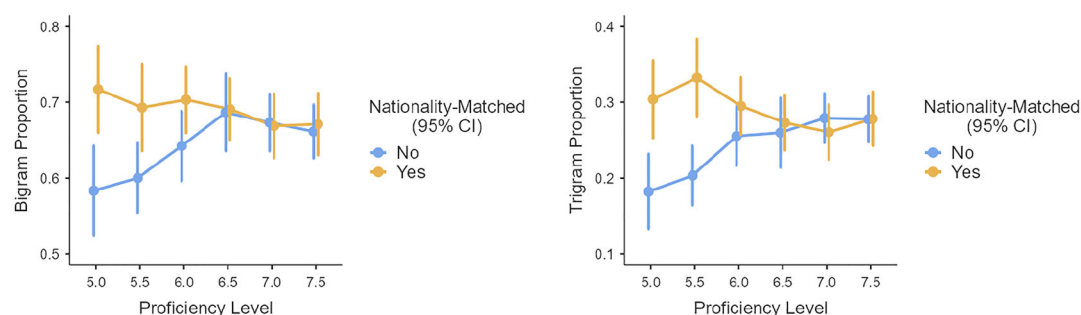


FIGURE 4 Effects of pair-level nationality on bigram and trigram proportions across proficiency levels. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1111/modl.70064)]

phraseological knowledge, can produce more target-like MWSs and then enjoy processing advantages, which in turn enhances the overall speaking processes (Boers et al., 2006; Skehan, 1998) and leads to a higher proficiency score. One plausible explanation for the lack of predictive power of MWS measures in this study concerns the difference in the choice of task modes: dialogic task (the current study) and monologic tasks (the previous studies). The differences between the use of MWSs in monologic and dialogic performance may be related to the inherent differences between the two task types, the differential aims and structures of the tasks, and the kinds of linguistic units required in each. Compared to a monologue, a dialogue is interactive by nature, typically divided into shorter and quicker turns (Michel et al., 2007), and is likely to offer fewer opportunities for the use of MWSs. In a monologue, the speaker's focus is on communicating a message with little or no opportunities for feedback, or a need to align one's speech with an interlocutor. In contrast, in a dialogue, the focus is on ensuring the view is expressed clearly to the interlocutor, the interlocutor's ideas are explored, a shared understanding is established, an agreement is reached, and/or a decision is made. During a conversation, speakers inherently accommodate their language and speech patterns to become more similar to their interlocutors, and as such, differences between interlocutors' linguistic choices reduce.

Relatedly, the cognitive processes underlying a phenomenon of linguistic alignment can account for the lack of a phraseology-speaking link in a dialogue. Alignment is known to occur as a result of output-input coordination (Garrod & Anderson, 1987). Specifically, the speaker formulates utterances using the same generation process (e.g., lexical, phonological, grammatical encoding) as is required when interpreting the recent relevant utterances produced by the interlocutor (Garrod & Anderson, 1987). From a psycholinguistic perspective of L2 speech production (Kormos, 2006), this indicates that background decisions and linguistic computations required for fluent speech production have already been executed in advance (Garrod, 1999), leaving little room for formulation processes to benefit from an additional processing advantage associated with the use of MWSs.

Why does nationality matter in the use of MWSs for lower-proficiency speakers?

The results of this study showed that nationality-matched speakers tended to produce a greater proportion of bigrams and trigrams than nationality-unmatched counterparts. More importantly, the impact of nationality was found to be more salient for lower-proficiency speakers (see Figure 4 above). Specifically, lower-proficiency speakers (5.0 and 5.5 levels) in a nationality-matched pair produced a greater number of target-like *n*-grams than those in a nationality-unmatched pair. However, the gap between the two groups became smaller as their proficiency levels increased up to 6.0 and 6.5 levels, and no substantial gap was observed for higher-proficiency levels (6.5 to 7.5 levels). Take the results of

Example 1: Kazakh Speakers

A (Lv.6.5): ... don't you think it will affect *young generation* like it it may, they may suffer cause of this policy *lack of* opportunities for them, *lack of* jobs.

B (Lv.5.5): Well, I think, uh, it won't be a problem because they're already facing this problem, the *lack of*, you know, the *younger generation* so ...

Example 2: Chinese Speakers

A (Lv.6.0): ... and some don't like arts? ur so uh *I agree with you* on the first point, but I have some different opinion different options on the second?

B (Lv.5.0): umm *I agree with you* about that point, umm but I think that art is er very common and ordinary to er to our our person, but ...

FIGURE 5 Examples of bigram and trigram alignments in nationality-matched pairs.

bigram proportion as an example, we found a significant difference only for the contrast between 5.0 and 6.5 ($b = 0.103$, $p = .010$). Yet, there was a trend of linearity between bigram proportion and proficiency level for unmatched nationality pairs (i.e., $M_{5.0} = 0.583$, $M_{5.5} = 0.600$, $M_{6.0} = 0.642$ and $M_{6.5} = 0.691$) compared to the results for matched nationality pairs (i.e., $M_{5.0} = 0.717$, $M_{5.5} = 0.693$, $M_{6.0} = 0.703$ and $M_{6.5} = 0.673$).

Why did the speakers in shared nationality pairs produce a greater proportion of n -grams? This can be explained based on accumulated evidence regarding the pair-level factors that are found to contribute to oral proficiency in a dialogue (e.g., Jaiyote, 2015; Van Engen et al., 2010) as well as linguistic alignment between the speaker and the conversation partner (e.g., Michel et al., 2022; Pickering & Garrod, 2021). Research on paired speaking assessment has shown that shared L1 speakers can enhance mutual understanding even when an utterance is unclear or ambiguous by drawing on multiple shared linguistic and non-linguistic resources, including their familiarity with the partner's accents (Harding, 2012), cultural background (Kachi, 2004), and knowledge of commonly occurring L1 driven grammatical errors (Jaiyote, 2015). Under this situation, the speaker is perhaps more likely to align with the partner in producing MWSs (see Figure 5 for two examples) because of the decreased processing demand on their cognitive capacity (Costa et al., 2008). Alignment may not occur as often for the speaker in a non-shared L1 pair, given that they would experience increased processing demand as a result of not being able to use as much background knowledge as in a shared L1 pair.

Crucially, the effect of nationality was particularly observed for lower-proficiency speakers (Levels 5.0 to 6.0). Perhaps it is the low-proficiency speakers who enjoy the greatest benefit from working with an interlocutor with the same L1 and cultural knowledge because they could use that knowledge to compensate for their lack of phraseological resources in formulating utterances. In contrast, the lower-proficiency speakers who were not paired with same-nationality interlocutors might have needed to rely more on their own phraseological knowledge, as indicated by the positive relationship between n -gram proportion and proficiency levels up to around 6.0 or 6.5 (see Figure 4 above).

Previous research (O'Sullivan, 2002) has provided evidence that acquaintanceship in a paired-speaking task can help interlocutors to achieve better proficiency scores and demonstrate language of higher syntactic complexity. In line with this body of research, our findings also suggest that low-proficiency speakers benefit from speaking to an interlocutor from the same nationality. In our study, however, it was not possible to investigate acquaintanceship independently, and as such, there could be an overlap between nationality and acquaintanceship, as some of the interlocutors might have known each other before their test performance. Future research will need to investigate these variables separately and in more depth.

CONCLUSION

The current research challenges the common view that the use of MWSs is a reliable indicator of higher oral proficiency, with our findings demonstrating that this is not the case when L2 speaking proficiency is measured via a dialogic task. This study was considered the first to revisit the central role of the ability to use MWSs, which has been documented based on monologic speaking performance, revealing that previous findings may not be extended to the case of dialogic speaking performance. We also found that one of the pair-level characteristics (i.e., nationality background) had a significant impact on the number of target-like MWSs used in dialogic speech. Furthermore, nationality significantly modulated the relationship between the use of MWSs (i.e., *n*-gram proportion) and L2 speaking proficiency. Our findings provide implications for L2 speaking assessment in a paired task format. It is important to note that the use of MWSs may not serve as a useful indicator of speaking proficiency when speech is elicited in a paired format. Under some specific conditions, phraseological measures may work to reveal general speaking proficiency in dialogue (e.g., lower-proficiency speakers in a nationality-unmatched pair). However, assessors need to remember that results of paired speaking performance are subject to multiple factors, including the individual- and pair-level characteristics; thus, objective measures that have been found to serve as a reliable predictor of speaking proficiency in monologic tasks may not function as well in a dialogic speaking setting. Our finding resonates with Gao and Sun (2025), who suggest different rating rubrics may be needed for monologic and dialogic task performance, a topic that requires further investigation.

Furthermore, these findings have significant implications for teaching and learning, particularly in multilingual contexts, as they provide evidence that the arrangement of dyads and pairwork needs careful attention. In classroom-based assessment contexts (e.g., diagnostic or placement purposes), language instructors would note that while the choice of interlocutor may not have a direct impact on the language used by proficient L2 speakers, lower-proficiency speakers may be disadvantaged in demonstrating their ability to use MWSs when paired with an interlocutor from a different nationality. As for L2 vocabulary instruction, our findings suggest that lower-proficiency learners (e.g., Level 5.0 up to around 6.0 in our case) may benefit more than higher proficiency learners from an MWS-focused instruction, particularly when they aim to engage in interactive communication with speakers with different L1 backgrounds.

Note, however, that our findings should not be generalized to different instructional or assessment contexts before more evidence is accrued in this line of research. Crucially, the scope of MWS measures in this study was limited to two and three consecutive strings of words while structural completeness (e.g., *on the other* vs. *in the end*), syntactic structures (e.g., verb + noun, adjective + noun), or complexity of sequences (e.g., phraseological diversity) were not considered. Furthermore, the relatively small sample sizes within each proficiency category ($n = 19\text{--}25$) should be acknowledged, and the present findings need to be replicated with a larger sample to verify the null relationship between MWS use and oral proficiency in dialogic performance. Another limitation is the use of nationality as a broad proxy for linguistic and non-linguistic characteristics, such as first language and cultural background. This approach may obscure within-nationality variation (e.g., differing first languages). Future research should more directly examine specific speaker-related variables (e.g., dominant language and acquaintanceship) in paired speaking assessments. Future research should also examine the extent to which the choice of reference corpus influences the results. Although this study employed the spoken section of COCA, which consists largely of scripted speech (e.g., television and radio programs), its suitability warrants further scrutiny, given that the present study focused on L2 interactional data produced in an assessment context (cf. student talk corpora; Dang & Dao, 2025). Finally, the effects of linguistic alignment warrant further investigation in future research. Although a comprehensive examination of alignment was beyond the scope of the present study, emerging patterns (e.g., pairwise correlations in bigram proportions) provided indirect evidence for its potential relevance. This issue should be explored more systematically in future studies using

quantitative measures of lexical or semantic overlap (e.g., Semantic Latent Similarity; see Y. Kim & Michel, 2023, for a methodological review), not only between test-takers but also between examiners and test-takers (Tantucci et al., 2025). We believe that the current study using general measures of MWSs may set the initial baseline to be compared to or challenged by future studies, which will provide further insights into the understanding of the relationship between phraseological competence and L2 speaking performance in real-life communicative settings.

AUTHOR CONTRIBUTIONS

Takumi Uchihara: conceptualization (equal); methodology (lead); writing—original draft preparation (equal); formal analysis (lead); writing—review and editing (equal). **Parvaneh Tavakoli:** conceptualization (equal); resources (supporting); writing—original draft preparation (equal); methodology (supporting); writing—review and editing (equal); project administration (lead). **Svetlana Mazhurnaya:** data curation (lead); resources (lead); writing—review and editing (equal). **Phil Smyth:** data curation (supporting); resources (supporting); writing—review and editing (equal).

DATA AVAILABILITY STATEMENT

The authors do not share the data.

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ENDNOTES

¹ Although it would have been ideal to examine additional pair-level variables (e.g., age, acquaintanceship and length of residence), only three relevant variables (gender, nationality and proficiency) were available for analysis. Restrictions related to test security and ethical considerations prevented test providers from sharing further participant information.

² TEEP classifies test-takers into predefined categories (i.e., Levels 0–9), each of which is described in detail through can-do statements. Raters are trained to use these established categories, along with detailed descriptors of expected performance at each level, when assigning ratings to individual aspects of the construct (e.g., fluency and accuracy). Accordingly, in this study, L2 proficiency was treated as a categorical variable in the analysis.

³ In comparison to CEFR levels, B1 proficiency may be situated within the lower-to-mid range of scores around 5.0–5.5, B2 proficiency within the mid-range around 6.0–6.5, and C1 proficiency within the higher range around 7.0–7.5; however, these alignments should be understood as approximate rather than as precise equivalencies.

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